PeshGo · پیشگو — dataset proposal

The full declaration is DATASET_DECLARATION.md in the repository.

QUESTION

What will the mean PM2.5 in Lahore be over the next 24 hours, how sure are I, and what should an outdoor worker, a parent or a school do about it?

WHY IT MATTERS

Lahore is regularly among the most polluted cities in the world from October to January. The decision (shift work, hold assembly indoors, send a child with asthma to school) is made the evening before, and advisories assume people can stay indoors. The US EPA guide lists outdoor workers among the groups most exposed.

SOURCES (REAL DATA ONLY)

Source	Use	Volume
OpenAQ v3: PM2.5 monitors within 25 km of Lahore	target and lags	543,283 hourly rows from 78 sensors
Open-Meteo: ERA5 weather archive and forecast	features	36,270 hourly rows
Open-Meteo: Copernicus CAMS air quality	feature, second baseline	36,270 hourly rows
NASA FIRMS VIIRS fires (not on the track list)	upwind fire counts	761 detections
Punjab EPD daily AQI reports	validation only	daily PDFs

Coverage 2022-09-02 to 2026-10-21. More than 50,000 target rows from OpenAQ alone.

TARGET, SPLIT AND BASELINE (FIXED IN CONFIG.YAML BEFORE ANY RESULT)

Target: mean PM2.5 of hours t+1 to t+24 per monitor, with at least 18 observed hours. Split by time, never shuffled: train to 31 Oct 2024, validation to 31 Mar 2025, test after, with a 24 h purge at each boundary. Declared baseline: persistence (the next 24 h equal the last 24 h); second reference: the CAMS forecast.

REPRODUCIBILITY

`make data` fetches every source, saves raw files unchanged with SHA-256 in `manifest.csv`, and rebuilds the cleaned and gold layers. Settings live in `config.yaml`; keys only in `.env`.

ETHICS

Public, non-personal data. Advice is information, not medical advice, and defers to Punjab government and school notices. No synthetic data is used for training or evaluation. Low-cost sensors are marked in humid hours, not corrected.